

Ahmed Bayoumy, PhD

Education

- 2019 **PhD**, *McGill University*, Montreal
Mechanical Engineering Department, Systems Optimization Lab
A Relative Adequacy Framework for Multi-model Management in Single- and Multi-disciplinary Design Optimization
- 2014 **MSc**, *Cairo University*, Egypt
Mechanical Design and Production Engineering
Modeling and Simulation of Large-Size Wind Turbine Blades Geometry Using Absolute Nodal Coordinate Formulation with Enhancement of its Performance
- 2008 **BEng**, *Benha University*, Egypt
Mechanical Engineering Department

Research Experience

- 2015–2019 **A Relative Adequacy Framework for Multimodel Management in Single- and Multidisciplinary Design Optimization**, *Research Assistant*, McGill, Montreal, Canada
- Formulate a relative adequacy framework (RAF) to address the trade-off between models' adequacy (usefulness) and their estimated computational cost in a design optimization context.
 - Accelerate the design optimization process utilizing the RAF while maintaining, or even improving, the trajectory of design performance.
 - Manage the selection, execution, and rectification of available computational models throughout the design optimization process.
 - Design two algorithms for solving multifidelity time-invariant multidisciplinary design optimization (MDO) problems using RAF.
 - Design an algorithm for solving multifidelity time-dependent MDO problems using RAF.
- 2018–2020 **Distributed Multidisciplinary Analysis and Design Optimization (DMADO)**, *NSERC CRD Research Assistant*, Siemens, McGill, and École de technologie supérieure (ETS), Montreal, Canada
- Develop a distributed multidisciplinary design optimization (DMDO) Python package to solve MDO problems using a penalty-based distributed interdisciplinary feasible (IDF) formulation, specifically the non-hierarchical analytical target cascading (NHATC) method.
 - Utilize the DMDO package to automate and accelerate the convergence of feedback coupling between multiple components.
 - Configure the MDO workflow for selected design components of aero-derivative gas turbine engines (AGT) using two MDO architectures: monolithic multidisciplinary feasible (MDF) and IDF approaches.
 - Integrate the developed DMDO package with Siemens software solutions, including Siemens SimCenter HEEDS.

- 2020-2023 **Improve algorithmic performance of Siemens SimCenter HEEDS solvers**, *Advanced Numerical Methods Engineer and Optimization Specialist*, Siemens DISW, Montreal, Canada
- Enhance the performance of algorithms for solving high-dimensional problems.
 - Accelerate design progress during the optimization process.
 - Design and implement a benchmarking suite to statistically evaluate algorithm performance and performance indicators on various single- and multi-objective design optimization problems under different random states.
- 2022-2023 **Distributed AutoML**, *Advanced Methods Engineer and Optimization Specialist*, Siemens DISW and McGill, Montreal, Canada
- Develop methods to solve the distributed combined algorithm selection and mixed hyperparameter optimization (CASH)
 - Cluster the configuration space to improve sampling efficiency.
 - Test the proposed methods on standard benchmark problems.
 - Apply the proposed approaches to uncertainty-aware AI models.
- 2023-2024 **Systems Multidisciplinary Analysis and Design Optimization**, *Advanced Numerical Methods Engineer and Optimization Specialist*, ([Solution Blog](#)), Siemens DISW, Montreal, Canada
- Adopt my open source [DMDO](#) Python package as a back-end to enable collaborative distributed optimization with SimCenter HEEDS tool.
 - Adopt the non-hierarchical analytical target cascade (NHATC) to solve distributed MDO problems.
 - Visualize, setup, and decompose MDO workflows using an interactive design structure matrix (DSM) front-end.
 - Enable design coordination in parallel execution mode.
- 2023-present **Uncertainty-aware AI-accelerators for Blackbox Optimization (BBO)**, *Siemens DISW*, Montreal, Canada
- Assess trustworthiness of the uncertainty-aware AI model predictions during the BBO process.
 - Leverage the design performance of best designs found without hurting the acceleration benefit of using AI models.
 - Break-off possible pre-mature convergence (plateau) of design performance when AI models predictions are excessively used to evaluate critical design candidates.
 - Use our proposed AutoML approach to manage the auto-selection and auto-tuning of AI models.
- 2024-present **AI-driven Systems MDO**, ([Mitacs Accelerate](#)), Siemens DISW and McGill, Montreal, Canada
- Proposal preparation with Prof. Kokkolaras
 - Data-driven systems design and optimization in distributed MDO architectures.
 - AI-based coordination accelerators.
 - Bayesian systems design and optimization.
 - Auto selection and tuning of models during the design process.

Employment History

- 2020–present **Advanced Software Engineer and Optimization Specialist**, *Siemens Digital Industries Software*, Montreal, Canada
- Contribute to developing innovative engineering design and optimization software solutions.
 - Implement cutting-edge technologies and research findings related to Multidisciplinary Design Optimization (MDO), artificial intelligence (AI), simulation-based optimization, and derivative-free optimization.
 - Propose and develop approaches, strategies and solutions for pressing technical challenges associated with engineering design optimization, AI and MDO such as:
 - Hierarchical and non-hierarchical MDO
 - Nonhierarchical coordination among distributed agents
 - AI-driven MDO
 - Automated machine learning (AutoML)
 - Uncertainty quantification
 - Multifidelity models
 - Knowledge-based optimization
 - Derivative free optimization
 - Reduced order models (ROMs) and data-driven dynamics
 - Influence studies
 - Benchmarking
 - Work in Agile environment; utilize Agile scrum practices.
- 2017–2020 **Research Assistant**, *Siemens Energy*, Montreal, Canada
- Algorithms design and development of distributed MDO architectures and optimization methods that reinforce decision-making in the design process of aeroderivative gas turbine engines.
 - Implement state-of-the-art technologies and research findings in process automation, simulation-based optimization and derivative-free optimization.
 - Integrate the developed methods into Siemens technology software packages (SimCenter and HEEDS):
 - AutoOpti and Moika
 - Process chain
 - ACES
 - NX open
 - Work in Agile environment; utilize Agile scrum practices.
- 2012–2015 **Engineering Group Leader**, *Power Generation Engineering and Services Company (PGESCO)*, *BECHTEL corporation*, Cairo, Egypt
- Developed a finite element analysis and design application software package; Pipe Supports Design and Analysis (PSD), based on linear structural analysis and international design codes and standards.
 - Contributed to the coordination and integration of combustion turbine and steam turbine packages for combined cycle plants and boiler-STG packages for thermal power plants, co-generation plants, and gas- and oil-fired plants.
 - Contributed to developing design guides for conducting structural, flow, and fluid-structure interaction analyses for critical systems at off-design conditions and different modes of plant operation, e.g., start-up, trip, and shut-down.
- 2010–2012 **Mechanical Design Engineer**, *EIE Group Company*, Cairo, Egypt
- Instructor for mechanical modeling and dynamic simulation using UNIGRAPHICS NX, AUTODESK INVENTOR, and ANSYS.
 - Reverse engineering; modeling using FARO laser scanning arm and GEOMAGIC.
 - Specialist of piping design and stress analysis.

Leadership and Supervision

- 2023–present **Supervisor and Mentor of Research Interns**, *Siemens DISW*, Montreal, Canada
- 2023–present **Research and Development Projects Scrum Master**, *Siemens DISW*, Montreal, Canada
- 2019–2020 **Research Group leader**, *Siemens*, Montreal, Canada

- 2014–2015 **Deputy Engineering Group Supervisor**, *PGESCO*, *BECHTEL* corporation, Cairo, Egypt, Hydraulic and Transient Analysis Group
- 2013–2014 **Plant Design Engineering Group Leader**, *PGESCO*, *BECHTEL* corporation, Cairo, Egypt, Stress Analysis Central Group
- 2012–2013 **Mechanical Engineering Group Leader**, *PGESCO*, *BECHTEL* corporation, Cairo, Egypt, Design Review Central Group

Awards

- 2015–2018 **McGill Engineering Doctoral Awards (MEDA)**, *McGill University*, Montreal, Canada
It is awarded in the amount of \$37,000 CAD each year for three academic years, a total value of \$111,000 CAD.

Articles in Archival Journals

- [1] A. Bayoumy and M. Kokkolaras. Multi-model Management for Time-dependent Multidisciplinary Design Optimization Problems. *Structural and Multidisciplinary Optimization*, 61(5):1821–1841, 2020.
- [2] A. Bayoumy and M. Kokkolaras. A Relative Adequacy Framework for Multimodel Management in Multidisciplinary Design Optimization. *Structural and Multidisciplinary Optimization*, 62(4):1701–1720, 2020.
- [3] A. Bayoumy and M. Kokkolaras. A Relative Adequacy Framework for Multi-Model Management in Design Optimization. *Journal of Mechanical Design*, 142(2), 2019.
- [4] A. Bayoumy, A. Nada, and S. Megahed. Methods of Modeling Slope Discontinuities in Large Size Wind Turbine Blades using Absolute Nodal Coordinate Formulation. *Proceedings of the Institution of Mechanical Engineers, Part K: Journal of Multi-body Dynamics*, 228(3):314–329, 2014.
- [5] A. Bayoumy, A. Nada, and S. Megahed. A Continuum Based Three-Dimensional Modeling of Wind Turbine Blades. *Journal of Computational and Nonlinear Dynamics*, 8(3), 2012.

Articles in Conference Proceedings

- [1] T. Peoc'h, A. Bayoumy, M. Staniszewski, H. Moustapha, M. Kokkolaras, and F. Garnier. Integration of Secondary Air System for Multidisciplinary Design Optimization of Gas Turbines. In *AERO2019*, Laval, Quebec, Canada, 2019. Canadian Aeronautics and Space Institute.
- [2] A. Bayoumy and M. Kokkolaras. A Relative Adequacy Framework for Multimodel Management in Multidisciplinary Design Optimization. In *Multidisciplinary Analysis and Optimization Conference*, Atlanta, Georgia, USA, 2018. AIAA.
- [3] A. Bayoumy and M. Kokkolaras. A Reference Error Formulation for Multi-fidelity Design Optimization. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, volume 58134, Cleveland, Ohio, USA, 2017. ASME.
- [4] A. Papadopoulos, M. Ismail, and A. Bayoumy. Dynamic Amplification Factor for Rigid and Flexible Piping System due to Steam Hammer Transient Load. In *ASME International Mechanical Engineering Congress and Exposition*, volume 57397, Houston, Texas, USA, 2015. ASME.

- [5] A. Bayoumy and A. Papadopoulos. Time History Steam Hammer Analysis for Critical Hot Lines in Thermal Power Plants. In *International Mechanical Engineering Congress and Exposition*, page 11, Montreal, Quebec, Canada, 2014. ASME.
- [6] A. Bayoumy, A. Nada, and S. Megahed. Use of Forward Dynamics Model for Designing Large-size Wind Turbine Blades. In *ASME International Mechanical Engineering Congress and Exposition*, volume 56253, San Diego, California, USA, 2013. ASME.
- [7] A. Bayoumy, A. Nada, and S. Megahed. Modeling Slope Discontinuity of Large Size Wind-turbine Blade Using Absolute Nodal Coordinate Formulation. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, volume 45059, pages 105–114, Chicago, Illinois, USA, 2012. ASME.

Workshop Talks and Guest Lectures

- 2022 **2nd AIAA Workshop for Multifidelity Modeling in Support of Design and Uncertainty Quantification**, AIAA, Chicago, USA, [Workshop Link](#)
- 2022 **MECH 559: Engineering Systems Optimization**, Guest lecture on Multidisciplinary Design Optimization, McGill University, Montreal, Canada, [Session Notebook](#)
- 2022-2024 **MECH 559: Engineering Systems Optimization**, Guest lecture on Derivative Free Optimization, McGill University, Montreal, Canada, [Slides Link](#)
- 2024 **MultiDisciplinary Design Optimization**, Learning Day Session, Siemens DISW, Montreal, Canada
- 2025 **Interview on the Role of Strategic AI in Engineering Design of Complex Systems**, Siemens Internal Communications, Siemens DISW, Lyon, France

Reviewer

- 2024-present **Scientific Reports**, Springer
- 2022-present **Journal of Computational Science**, Springer
- 2020-present **ASME Journal of Mechanical Design**
- 2020-present **Structural and Multidisciplinary Optimization journal**, Springer
- 2015-present **Journal of Multi-body Dynamics**, SAGE

Teaching Assistant

- 2016, 2019 **MECH 559: Engineering Systems Optimization**, McGill University, Montreal, Canada
- 2017, 2019 **MECH 501, 502: Analysis, Synthesis and Optimization of Engineering Systems**, McGill University, Montreal, Canada
- 2017-2019 **MECH 290: Design Graphics for Mechanical Engineering using SolidWorks**, McGill University, Montreal, Canada
- 2016, 2018 **MECH 292: Conceptual Design**, McGill University, Montreal, Canada
- 2016 **MECH 539: Computational Aerodynamics**, McGill University, Montreal, Canada
- 2018-2019 **MECH 400: Engineering Professional Practice**, McGill University, Montreal, Canada

Open source code library

I participated in the development of academic and industrial software projects that help reinforce engineering decision making in complex systems. I authored open-source codes that have been used in various research projects. Including, but not limited to, five python packages that can work together to manage the use of multiple models during design optimization. Each developed package can work per se to provide its function in other design exploration and exploitation contexts. The codes are maintained on the following GitHub repositories:

- Single- and Multi-Objective Derivative-Free Blackbox Optimization Package: Optimization using Mesh Adaptive Direct Search (OMADS) [[Link](#)] [[OMADS Documentation](#)]
- Distributed Multidisciplinary Design Optimization (DMDO) [[Link](#)] [[DMDO Documentation](#)]
- Surrogate Models Library (SML) [[Link](#)]
- Samplers Library (samplersLib) [[Link](#)]
- Relative Adequacy Framework (RAF) [[Link](#)]

Languages

English	Fluent
French	Basic
Arabic	Native